9. [Maximum mark: 7]

The sum of the first n terms of a geometric sequence is given by $S_n = \sum_{r=0}^{n-1} 5\left(\log_2 c\right)^r$.

(a) Given that S_n converges, find the range of possible values of c. [3]

(b) In the case where $c = 1.5$, find the least value of n such that $\left|S_\infty - S_n\right| < 0.1$. [4]